MIND MATRIX

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SESSION: 02

REPORT ON: LAB

TITLE: PROJECT LAB

PROJECT LAB:

This Kotlin Project Lab simulates the basic structure of a modern Android app using plain Kotlin to demonstrate how Jetpack Compose works conceptually. It recreates the flow of MainActivity, setContent {}, and @Composable functions to build a simple Greeting Card application in a console environment. The purpose of this lab is to strengthen understanding of declarative UI principles and prepare for real Android development using Jetpack Compose.

Kotlin Lab Code:

```
// ============================================================
// KOTLIN PROJECT LAB
// Topic: Simulating Your First Android App in Kotlin
// ============================================================


@Target(AnnotationTarget.FUNCTION)
annotation class Composable


// Simulating the MainActivity structure
class MainActivity {


    fun onCreate() {
        println("Launching Greeting Card Simulation...")


        // TODO 1 Completed
        setContent {
            GreetingScreen("Sahil")
        }
```

```
    }

    // Simulated setContent() method
    private fun setContent(content: () -> Unit) {
        println("Setting up the content...")
        content()
    }
}

// TODO 2 Completed
@Composable
fun GreetingScreen(name: String) {
    println("Hello $name! Welcome to your first Kotlin App!")
}

// TODO 3 Completed
@Composable
fun AppTheme(content: () -> Unit) {
    println("Applying GreetingCard Theme...")
    content()
}

// TODO 4 Completed
fun preview() {
    println("--- Preview Mode ---")
    GreetingScreen("Android Learner")
    println("--- End of Preview ---")
```

```
}

// ============================================================
// MAIN FUNCTION – entry point for simulation
// ============================================================

fun main() {

    println("========== Simulated Android App ==========\n")

    val activity = MainActivity()
    activity.onCreate()

    println()

    // TODO 5 Completed
    AppTheme {
        GreetingScreen("Sahil")
    }

    println()

    // TODO 6 Completed
    preview()

    println("\n========== End of Simulation ==========")
}
```